

Loop Redesign of TIMP3 for Selective Binding of MMP9 Over MMP2

Ryan Gustafson^{1,2}, Gavin Gallagher¹, Maryam Raeeszadeh-Sarmazdeh¹

¹Chemical and Materials Engineering, University of Nevada, Reno ²Integrated Neuroscience Program, University of Nevada, Reno



University of Nevada, Reno

Background

MMP2 and MMP9 are gelatinases with highly similar catalytic domains. MMP9 is associated with tumor invasion and MMP2 with basal basement-membrane maintenance, and broad-spectrum MMP inhibitors have not succeeded clinically, which motivates protease-selective binders [1-3]. How a binder discriminates between two nearly identical active sites is a structural question in its own right.

Tissue inhibitor of metalloproteinases 3 (TIMP3) inhibits both proteases by inserting an N-terminal ridge, formed by the AB, C, and connector loops, into the active-site cleft [4,5]. These loops were redesigned with generative models to test whether loop sequence and geometry alone can shift a scaffold's preference between homologous targets, and whether predicted interface metrics account for the preference that is measured.

Design Pipeline

Loop backbones were generated on the TIMP3 scaffold with RFdiffusion [6] (insertions of 6 to 15 residues in the AB and C loops, 4 to 10 in the EF loop; 25 backbones per target). ProteinMPNN [7] sampled sequences for each backbone, and candidates were co-folded with AlphaFold3 [8] against MMP2, MMP3, MMP9, and ADAM17.

Each variant was scored against its own cross-target mean (a self-normalized T-score), so that target-preferential variants rank above variants that score well on every target. From 549 co-fold results and 128 unique loop variants, 39 consensus candidates were identified and 15 were chosen by manual curation for synthesis.

Screening and Antigen Matching

Constructs were displayed on EBY100 yeast and analyzed by flow cytometry. Display level was read from an epitope-tag stain (FITC) and target binding from APC fluorescence. The primary readout was the median APC/FITC ratio among expressing cells (Pos Med Ratio), which normalizes binding to display level. MMP9 and MMP2 signals were compared with a two-sample Welch t-test.

The three antigen suppliers do not provide equivalent molecules. Pooling trials across suppliers would compare different proteins under one target label, so selectivity was tested only on Enzo material, which supplies human catalytic-domain constructs for both targets.

Supplier	MMP2	MMP9
Enzo	human, catalytic domain	human, catalytic domain
Sino	mouse, lacks N-catalytic half	human, full length
Masoud (in-house)	not supplied	boundaries undocumented

Conclusions

Redesign of the TIMP3 loops produced five variants with higher MMP9 than MMP2 signal on matched antigen, consistent in every replicate. The evidence is limited to two independent preparations per group measured on one day, so the fold differences and p-values are provisional. Predicted interface confidence did not explain the target-specific component of binding, which was small relative to the construct-level component. A structural feature separating MMP9 from MMP2 binding was identified only for C 12.

Next Steps

Replication on separate days with independent preparations will test whether the five differences persist and estimate day-to-day variance. Surface plasmon resonance against purified, construct-matched MMP9 and MMP2 will measure dissociation constants and separate affinity from display-dependent avidity. A second design generation will score off-target binding during selection, and a sequence-based classifier (fine-tuned ESM-C) will be tested against the new measurements.

Design and Screening Workflow

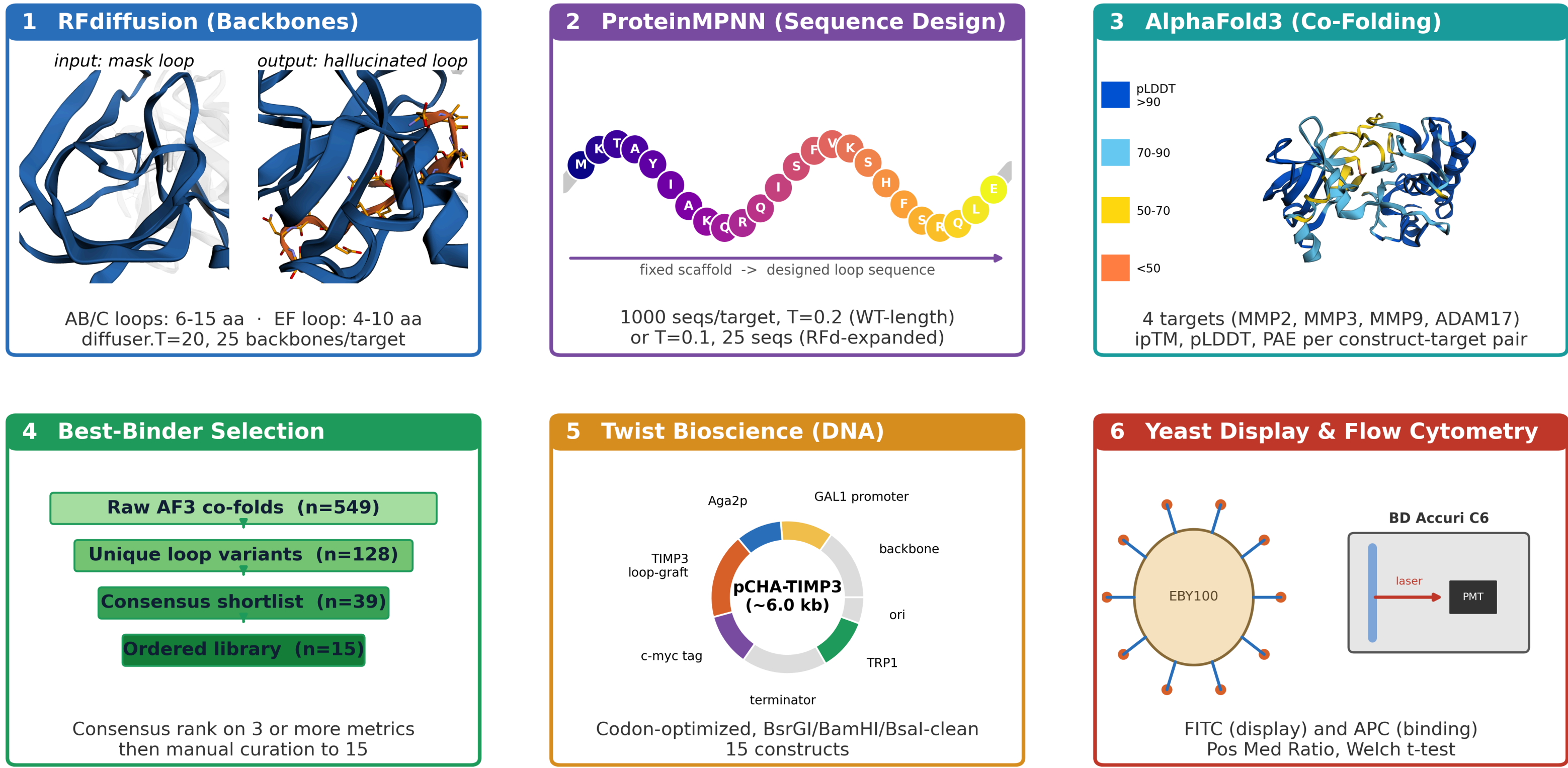


Figure 1. Design and screening workflow. Panel 1 shows the TIMP3-WT C loop masked as the RFdiffusion input (left) and the C 12 redesigned loop as output (right); panel 3 shows the C 12 co-fold with MMP9 colored by pLDDT.

Selectivity for MMP9 Over MMP2

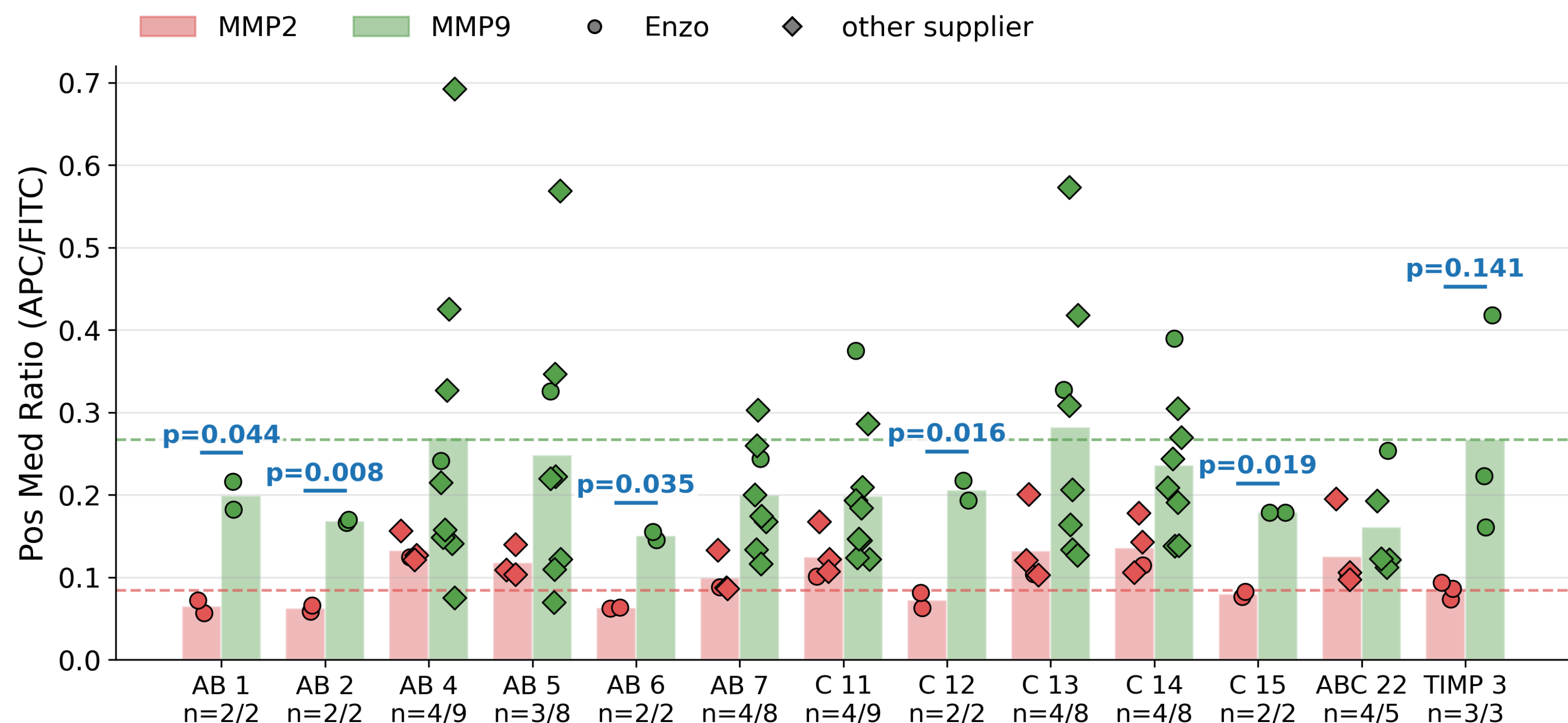


Figure 2. Pos Med Ratio, one point per independent preparation; bars are means and n (MMP2/MMP9) is under each label. Dashed lines are TIMP3-WT means. Brackets give nominal, uncorrected Welch p-values for Enzo-only constructs; other constructs are pooled across suppliers and untested.

All five variants designed for MMP9 preference (AB 1, AB 2, AB 6, C 12, C 15) had higher MMP9 than MMP2 signal in every replicate, with 2.2 to 3.1-fold differences. Each group contains two independent cultures with separate label preparations, all measured on one day. With five comparisons, only AB 2 falls below a Bonferroni threshold of 0.01, and on the threshold-based Binding Efficiency metric only AB 2 and AB 6 reached significance.

TIMP3-WT was not significant ($p = 0.141$, $n = 3$). Its MMP9 values were 0.161 and 0.223 on the first day and 0.418 on a second day, so between-day variation in a single construct can be as large as the MMP9 to MMP2 differences reported here.

Construct	MMP2	MMP9	Fold	Welch p
AB 1	0.065	0.199	3.1	0.044
AB 2	0.062	0.168	2.7	0.008
AB 6	0.063	0.150	2.4	0.035
C 12	0.072	0.206	2.8	0.016
C 15	0.079	0.179	2.2	0.019
TIMP3-WT	0.084	0.267	3.2	0.141

Enzo material only. Mean Pos Med Ratio; $n = 2$ per group (3 for TIMP3-WT).

Interface Geometry

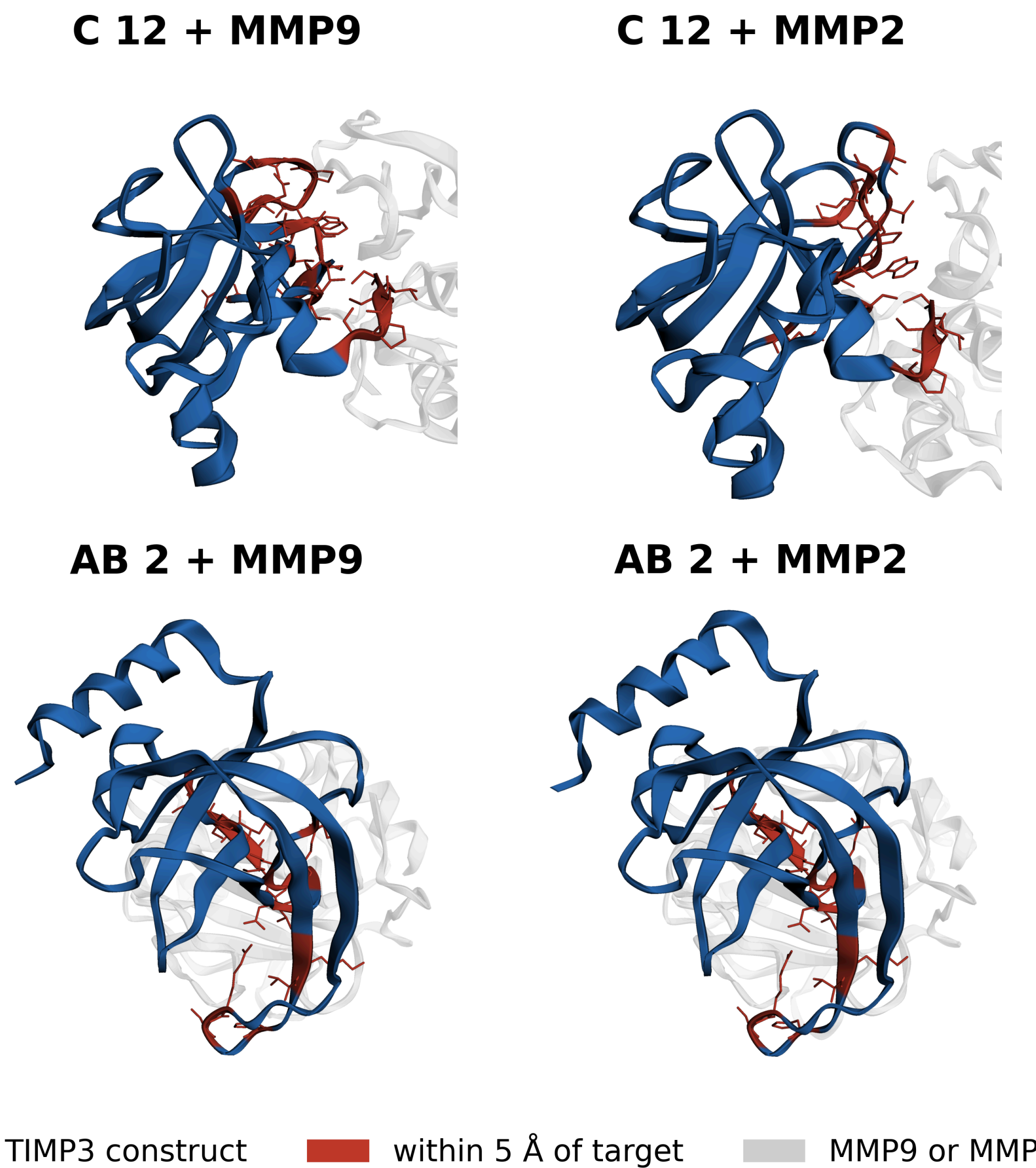


Figure 3. TIMP3-chain residues within 5 Å of the target (red) in AlphaFold3 co-folds. Each MMP2 co-fold is superimposed on its MMP9 co-fold by TIMP3-chain C-alpha atoms (RMSD 5.27 Å for C 12, 0.57 Å for AB 2).

Shape and charge complementarity, catalytic-loop occlusion, and buried surface area were compared between each variant's MMP9 and MMP2 co-folds. Shape complementarity differed by 0.03 or less for all five variants, and no single feature tracked selectivity across them.

C 12 was the exception. Its interface contained 26 TIMP3-chain residues at MMP9 and 18 at MMP2, with a buried-surface-area difference of +180 Å², and its redesigned loop adopted a different conformation against each target. AB 2 contacted 19 and 18 residues with the same loop conformation. These are single static predictions and do not distinguish a structural effect from binding kinetics or avidity.

Calibration of Predicted Metrics

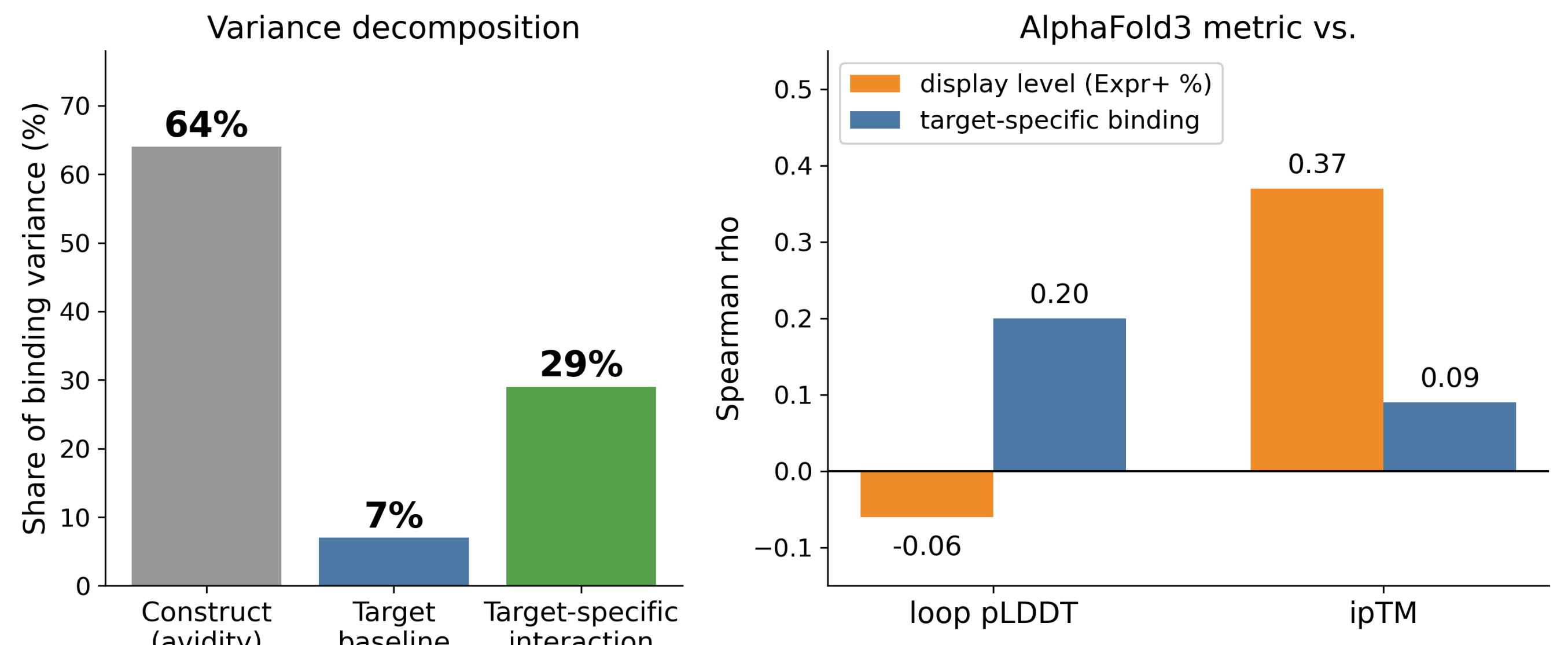


Figure 4. Left: variance decomposition of binding (12 constructs, 3 targets). Right: Spearman correlation of AlphaFold3 metrics with display level and with the target-specific residual ($n = 36$ construct-target pairs).

A construct-level component, which reflects general binding strength across targets, accounted for 64% of the variance in binding, the target baseline for 7%, and the target-specific interaction for 29%. After the construct component was removed, AlphaFold3 metrics did not correlate significantly with target-specific binding (ipTM $\rho = 0.09$, $p = 0.58$; loop pLDDT $\rho = 0.20$, $p = 0.23$). ipTM did correlate with display level ($\rho = 0.37$, $p = 0.027$).

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